

Ruizhi Cheng

CONTACT INFORMATION
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INDUSTRIAL EXPERIENCE
Meta Platforms, Inc., Menlo Park, CA
Research Intern/Collaborator Jul. 2025 - Present

- Proposed learning-based edge relay selection for Meta’s real-time communication products (*e.g.*, WhatsApp), leveraging real-time network-quality telemetry to optimize path selection in dynamic networks.
- Designed an end-to-end networked testbed for real-time human–GenAI interactions to support internal prototyping and product evaluation.
- Conducted large-scale measurement studies of commercial human–GenAI audio/video interaction applications, translating empirical findings into system design insights for Meta product development.
- Published 2 papers; 2 additional papers under review.

Nokia Bell Labs, New Providence, NJ
Research Intern Jun. 2024 - Aug. 2024

- Designed loss-resilient immersive telepresence systems.

EDUCATION
Ph.D. Candidate in Computer Science Aug. 2021 - Present
George Mason University Fairfax, VA

- Designed learning-based immersive networked systems for human–human and human–GenAI interaction.
- Expected Graduation: May 2026.

Bachelor of Science in Computer Science Sep. 2017 - Jun. 2021
Jilin University Jilin, China

TECHNICAL SKILLS
Domains: Networked Multimedia Systems; Real-time Communication (RTC); Network Measurement & Performance Analysis; Immersive 3D Content Streaming; Neural Rendering (NeRF/3DGS); ML Systems.
Languages/Frameworks: C++, Python, PyTorch, WebRTC, Media over QUIC (MoQ), Unreal Engine.

SELECTED PROJECTS
Optimizing Network Infrastructure for Real-time Human–GenAI Audio/Video Calling
(*Work done at Meta*)

- Proposed an asymmetric transmission paradigm leveraging MoQ and edge-based network adaptation to serve the unique traffic patterns of Human-to-GenAI calling.
- Designed an end-to-end networked testbed for GenAI calling supporting multiple transmission paradigms (*e.g.*, WebRTC and MoQ) and adaptation algorithms (*e.g.*, congestion control and bitrate adaptation).

Characterizing Video-based Conversational Interaction with Generative AI
(*Work done at Meta*)

- Conducted the first large-scale measurement study of video-based human–GenAI interaction applications.
- Built a fully automated measurement pipeline to run end-to-end interactions at scale (>10,000 sessions), quantifying conversational latency and backend vision–language model (VLM) inference accuracy, as well as analyzing transport/application-layer behavior and media adaptation (resolution/FPS/bitrate).
- Derived design insights such as conventional bitrate adaptation being misaligned with VLM consumers, and transport protocol choices substantially affecting system robustness under packet loss.

Bandwidth-conserving Immersive Telepresence via Semantic Communication

- Built an ultra-low-bandwidth immersive telepresence (live, interactive volumetric video) system leveraging semantic communication.
- Designed techniques to extract and transmit compact motion and color semantics from RGB-D inputs, combined with an optimized neural rendering pipeline for realistic immersive content reconstruction.
- Operated at only 0.2 Mbps (a 1200× reduction compared to state-of-the-art) while maintaining high visual quality (0.92 SSIM), low end-to-end latency (95 ms), real-time streaming at 30+ FPS, and improving quality of experience (QoE) by up to 86.7%.

Loss-resilient Immersive Telepresence with Multimodal Semantics

(Work done at Bell Labs)

- Built a loss-resilient immersive telepresence system that recovers missing content at the semantic level within strict frame deadlines.
- Designed a lightweight multimodal model to recover lost motion semantics, and introduced *neural redundancy* to protect audio semantics with minimal communication and computation overhead.
- Improved QoE by up to 91% over existing schemes while sustaining real-time streaming at 30 FPS and achieving strong visual quality under real-world packet loss.

Measurement of Real-time Human–GenAI Audio Calling Applications

(Work done at Meta)

- Conducted an in-depth measurement study of commercial human–GenAI audio calling applications.
- Demonstrated that human–GenAI conversational systems require rethinking media delivery architectures: RTC-style low-latency streaming on the uplink and CDN-style scalable delivery on the downlink.
- Identified infrastructure bottlenecks, including excessive connection setup overhead in WebRTC-based systems and increased latency under batch-based delivery.

Privacy-preserving User Authentication in Virtual Reality with Federated Learning

- Designed a multimodal federated learning system for privacy-preserving virtual reality user authentication using behavioral biometrics, enabling on-device inference without sharing raw data.
- Achieved 28% higher accuracy and 76.2% lower communication overhead than baselines with <250 ms on-device inference time.

Advancing Volumetric Video Streaming with Neural Content Representation

- Designed a low-bandwidth, loss-resilient volumetric video streaming system that delivers neural content representations (*e.g.*, Neural Radiance Fields, NeRF) to mobile XR headsets.
- Proposed visibility-aware selection of NeRF feature voxels and learning-based voxel recovery to mitigate packet loss, reducing bandwidth by 68.3% while maintaining high visual quality under packet loss.

SELECTED
PUBLICATIONS

Publications marked with ★ indicate acceptance at top-tier conferences (per CSRankings).

8. **Ruizhi Cheng**, Surendra Pathak, Guowu Xie, Matteo Varvello, Songqing Chen, Bo Han
Hello, GenAI? Dissecting Human to Generative AI Calling
ACM IMC (★), 2025.
7. Nan Wu, Bo Chen, **Ruizhi Cheng**, Klara Nahrstedt, Bo Han
NeVo: Advancing Volumetric Video Streaming with Neural Content Representation
ACM MobiCom (★), 2025.
6. Nan Wu*, Weikai Lin*, **Ruizhi Cheng***, Bo Chen, Yuhao Zhu, Klara Nahrstedt, Bo Han
Advancing Immersive Content Delivery with Dynamic 3D Gaussian Splatting
ACM HotMobile, 2025. *: Equal contribution.
5. **Ruizhi Cheng**, Nan Wu, Vu Le, Eugene Chai, Matteo Varvello, Bo Han
MagicStream: Bandwidth-conserving Immersive Telepresence via Semantic Communication
ACM SenSys (★), 2024. **Best Paper Runner-up Award**
4. **Ruizhi Cheng**, Yuetong Wu, Ashish Kundu, Hugo Latapie, Myungjin Lee, Songqing Chen, Bo Han
MetaFL: Privacy-preserving User Authentication in Virtual Reality with Federated Learning
ACM SenSys (★), 2024.
3. **Ruizhi Cheng**, Nan Wu, Matteo Varvello, Eugene Chai, Songqing Chen, Bo Han
A First Look at Immersive Telepresence on Apple Vision Pro
ACM IMC (★), 2024.
2. **Ruizhi Cheng**, Erdem Murat, Lap-Fai Yu, Songqing Chen, Bo Han
Understanding Online Education in Metaverse: Systems and User Experience Perspectives
IEEE VR (★), 2024.
1. **Ruizhi Cheng**, Nan Wu, Matteo Varvello, Songqing Chen, Bo Han
Are We Ready for Metaverse? A Measurement Study of Social Virtual Reality Platforms
ACM IMC (★), 2022.